KOREA WEEKLY KICKSTART

KOSPI declined further by 2% amid continued risk-off sentiment across global AI names ahead of major earnings reports

Timothy Moe, CFA
+65-6889-1199 | timothy.moe@gs.com
Goldman Sachs (Singapore) Pte

John Kwon
+65-6654-6337 | jongmin.kwon@gs.com
Goldman Sachs (Singapore) Pte

- KOSPI declined further by 2% amid continued risk-off sentiment across global AI names ahead of major earnings reports. The Pharmaceutical, Software and Insurance sectors outperformed this week, while the Chemicals, Steel and Shipbuilding sectors underperformed the most (Exhibit 20).
- Foreign investors continued to sell the KOSPI market, driven by outflows for KOSPI Tech (Exhibit 47).
- KOSPI 12m-forward EPS was revised up by 0.1%. The Shipbuilding sector saw the strongest upward earnings revisions, while the Utilities sector was revised down the most this week (Exhibit 34).
- The KRW strengthened 0.7% vs. USD this week. Also, it strengthened 1.0% vs. EUR but weakened 0.3% vs. JPY.
- The latest Korea Equity Risk Barometer (GSSRKERB Index) is at -1.1, staying in a risk-averse territory.

Charts of the Week: Favorable risk/reward even with more conservative valuation assumptions; Hedge Fund and MF Positioning Pulse Check

- Korea's headline KOSPI index has fallen 40% and retraced half its gains over the prior bull run. The decline has been driven by investor concerns over the heavyweight memory sector and amplified by retail derisking. The sharp market drop in the context of record corporate profits has brought the 12m forward P/E to 4x (-3 s.d.). We examine memory and non-memory sectors along with volatility and earnings momentum and conclude the market is pricing an overly negative fundamental outlook. **We lower our KOSPI target from 12000 to 9000 to incorporate more conservative assumptions about how the market may price underlying fundamentals but still view the market as offering attractive risk/reward at current levels**. On these metrics, the market would be over 20% undervalued and would price elevated volatility and slowing earnings momentum. The Korean equity market appears deeply undervalued and offers over 60% upside from current levels. See more: Favorable risk/reward even with more conservative valuation assumptions.
- **Hedge Fund and MF Positioning Pulse Check:** According to GS Prime Services, the region remains net sold MTD, led by Taiwan, Korea, Japan and China; partially offset by net buying in Australia and Hong Kong. Korea Net allocation declined to 3.5% (31st percentile 1-year, 86th percentile 5-year), while gross

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

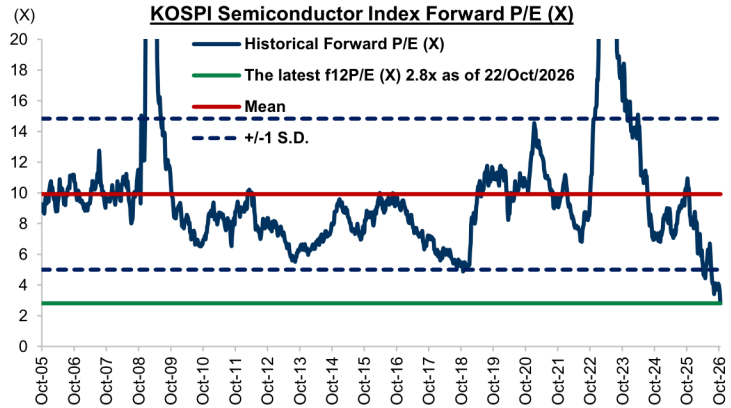
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allocation declined to 2.5% (35th percentile 1-year, 87th percentile 5-year). Based on preliminary EPFR data (~55% of total AUM reported), Asian funds are still UW with Korea despite increased exposures over the past 1/3 months (vs. benchmark).

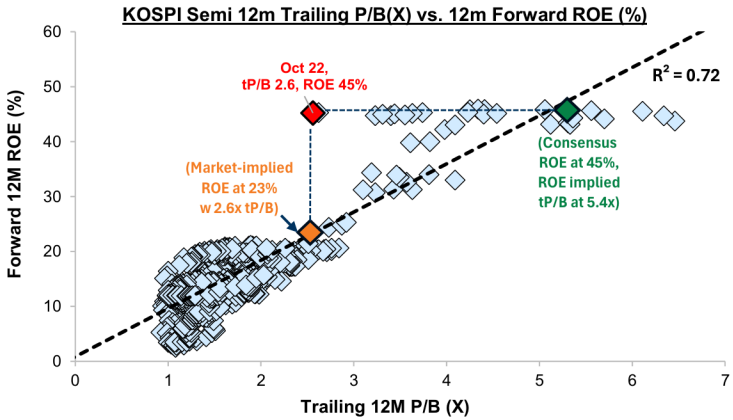
Charts of the week: Favorable risk/reward even with more conservative valuation assumptions

Exhibit 1: Korea's semiconductor memory sector is trading at record low forward P/E levels, indicating the market's lack of conviction in the sustainability of earnings



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 2: 12m forward ROE explains 72% of the variance in the semiconductor memory sector's trailing price/book valuation



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 3: Korea semiconductor sector 2027 earnings scenarios and resulting ROEs, implied PBR levels, and potential returns from current valuation

KOSPI Semi Sector Earnings Scenarios

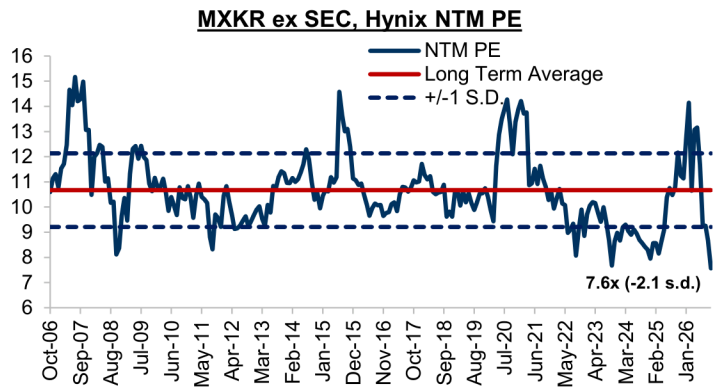
Potential f12m Earnings Revisions	-90%	-85%	-80%	-70%	-65% (Current LTM PBR)	-50%	-40%	-30%	-20%	-10%	0% (Current Earnings)	10%
Forward 12 month Book Value Revisions	-38%	-36%	-34%	-30%	**-28%**	-21%	-17%	-13%	-8%	-4%	**0%**	4%
Potential ROE for f12m (%)	8%	11%	15%	21%	**23%**	31%	35%	39%	42%	45%	**48%**	51%
Potential OP Margin for f12m (%)	7% (-0.9 s.d)	12% (-0.5 s.d)	16% (-0.1 s.d)	24% (+0.7 s.d)	**27%** (+1.0 s.d)	37% (+2.0 s.d)	42% (+2.5 s.d)	47% (+3. s.d)	52% (+3.4 s.d)	56% (+3.8 s.d)	**60%** (+4.2 s.d)	63% (+4.5 s.d)
ROE implied LTM PBR (A)	0.8	1.2	1.6	2.3	**2.6**	3.4	3.9	4.3	4.7	5.1	**5.4**	5.7
f12/t12 month Book Growth (B)	7%	11%	15%	22%	**26%**	37%	45%	52%	60%	67%	**75%**	82%
Re-Rating Upside (A) (Current LTM PBR 2.6x) + f12m Book Growth (B)	-63%	-44%	-27%	6%	**22%**	64%	89%	113%	135%	156%	**176%**	194%

**** Note : (1) Assuming dividends should be revised in the same magnitude of earnings revisions***

(2) The latest pricing is as of 22st of October, 2026

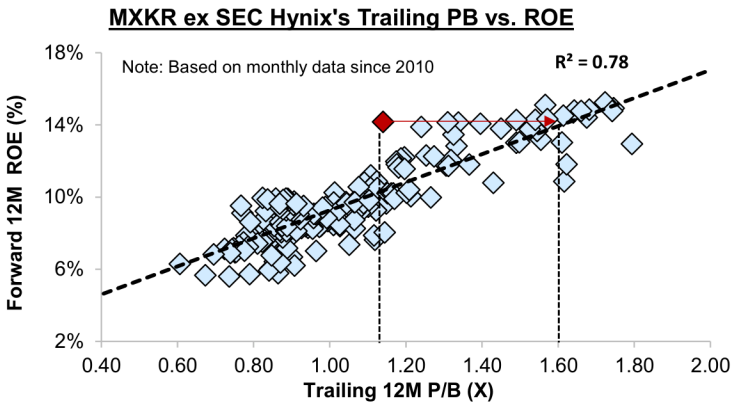
Source: Quantiwise, FactSet, Goldman Sachs Global Investment Research

Exhibit 4: Korea ex-memory stocks currently trade at a record low 7x NTM P/E



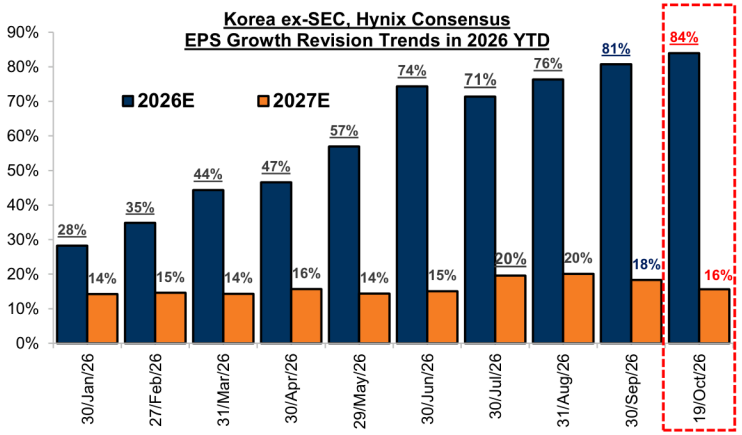
Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 5: The non-memory sector is more than 30% undervalued based on the strong historical relationship between forward ROE and trailing P/B



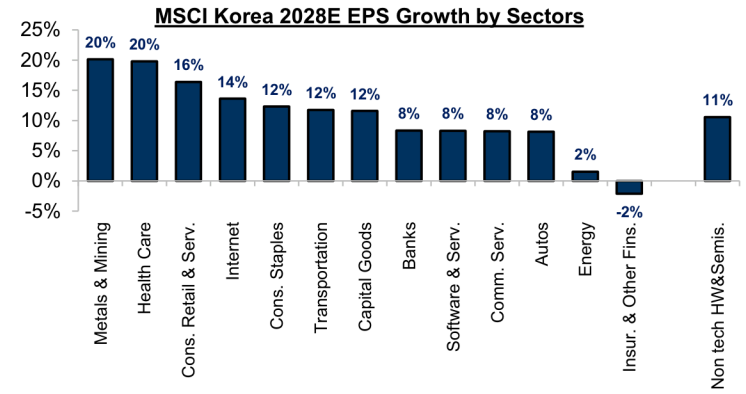
Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 6: Consensus growth expectations for the non-memory part of the market are strong for 2026 and 2027...



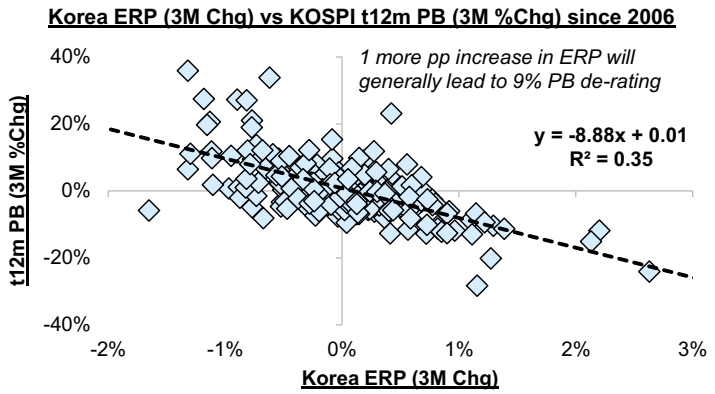
Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 7: ... with a further 11% increase in 2028



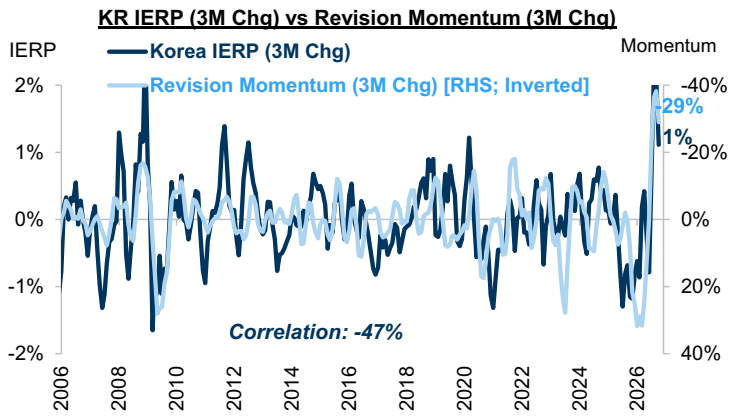
Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 8: A 1pp increase in Korea's ERP is roughly equivalent to a 9% de-rating in PBR



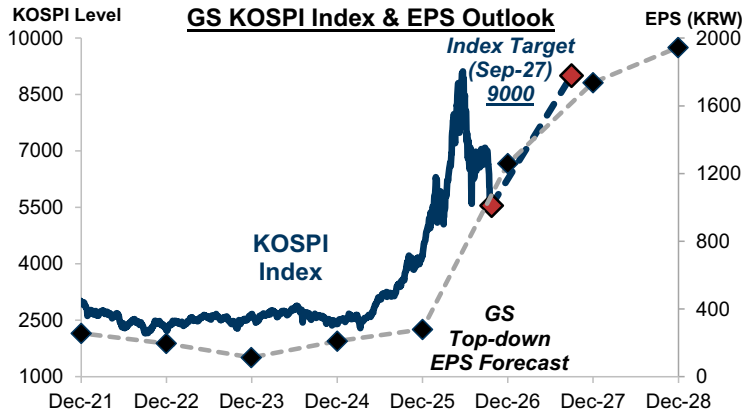
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 9: Earnings revision momentum correlate well with market valuation



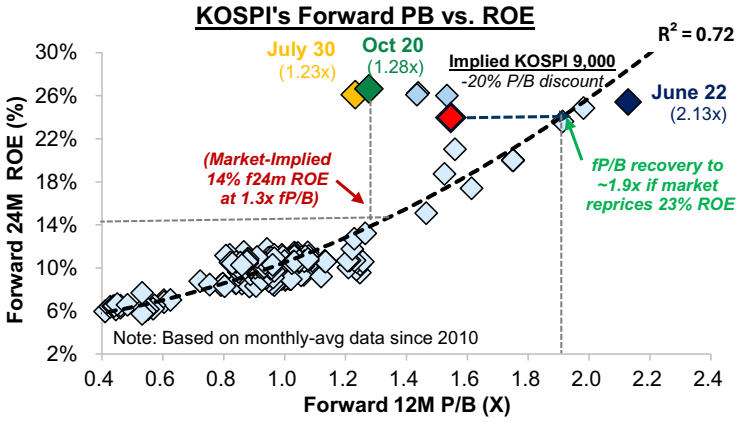
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 10: Korean equities have sharply undershot the aggregate earnings trajectory and have room to reboun



Source: Goldman Sachs Global Investment Research

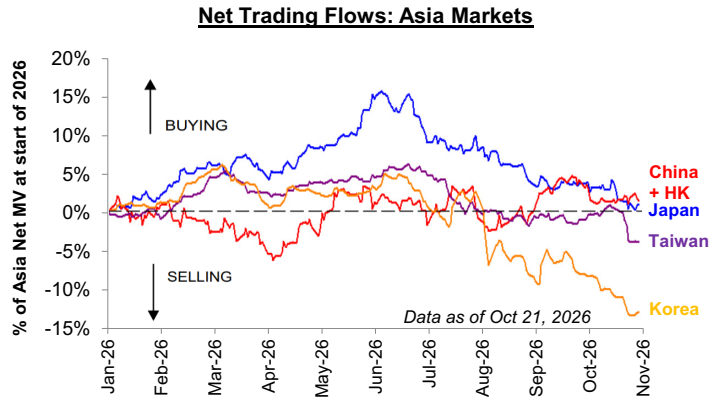
Exhibit 11: At our 9000 12m target, KOSPI would be over 20% undervalued based on its PB vs. ROE relationship



Source: FactSet, Goldman Sachs Global Investment Research

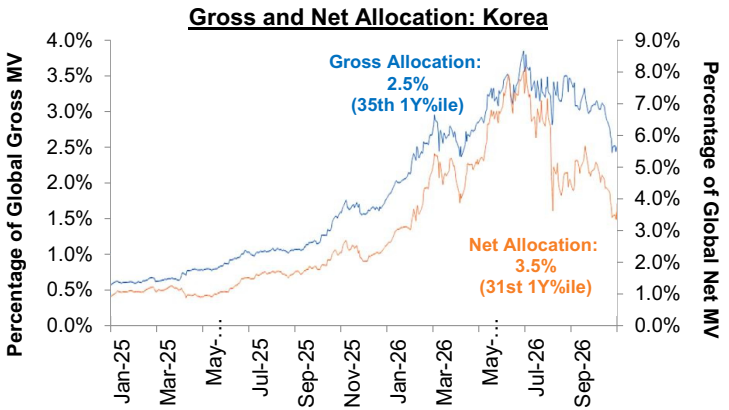
Charts of the week: Hedge Fund and MF Positioning Pulse Check

Exhibit 12: According to GS Prime Services, The region remains net sold MTD, led by Taiwan, Korea, Japan and China; partially offset by net buying in Australia and Hong Kong.
As of October 21, 2026



Source: Goldman Sachs FICC and Equities

Exhibit 13: Korea Net allocation declined to 3.5% (31st percentile 1-year, 86th percentile 5-year), while gross allocation declined to 2.5% (35th percentile 1-year, 87th percentile 5-year).



Source: Goldman Sachs FICC and Equities

Exhibit 14: Based on preliminary EPFR data (~55% of total AUM reported), Asian funds are still UW with Korea despite increased exposures over the past 1/3 months

AEJ funds (US$90 bn - Active Fund Only)				
Market	OW/UW allocation (in bp)	Market	1-mth chg	3-mth chg
China	230	Korea	25	140
Singapore	140	Taiwan	20	(55)
Hong Kong	125	Thailand	10	10
Indonesia	70	Malaysia	5	10
Thailand	60	India	5	60
Philippines	20	Indonesia	(1)	(15)
Malaysia	-35	Philippines	(1)	10
Korea	-65	Singapore	(10)	(15)
India	-215	Hong Kong	(20)	(35)
Taiwan	-490	China	(70)	(75)

Note: The allocation of AEJ mandate only includes active funds, as of Sep '26

Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 15: EM funds are most OW Brazil/Mexico, and most UW Taiwan. Over the past month, EM funds have raised exposure the most in India, and trimmed exposure in Taiwan

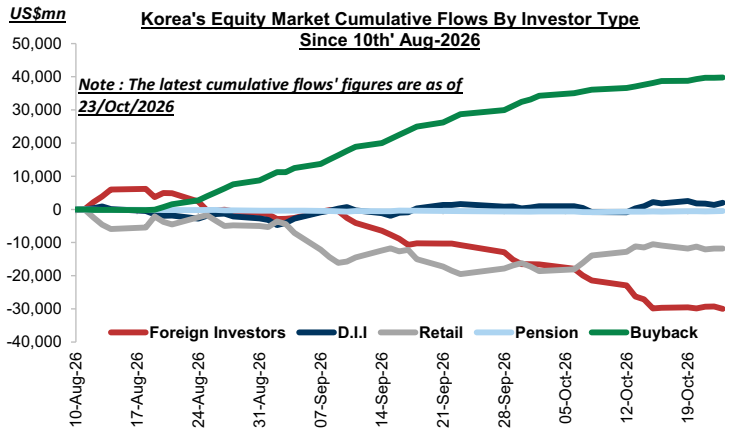
EM funds (US$220 bn - Active Fund Only)				
Market	OW/UW allocation (in bp)	Market	1-mth chg	3-mth chg
Brazil	275	India	15	45
Mexico	80	Saudi Arabia	10	10
Hungary	35	Brazil	10	10
Greece	30	South Africa	10	(10)
Indonesia	20	Malaysia	5	5
China	-45	Hungary	(5)	0
Saudi Arabia	-120	Turkey	(5)	(5)
India	-160	Mexico	(5)	2
Korea	-190	China	(15)	(40)
Taiwan	-460	Taiwan	(40)	(85)

Note: The allocation of GEM mandates only include active funds, as of Sep '26

Source: EPFR, Goldman Sachs Global Investment Research

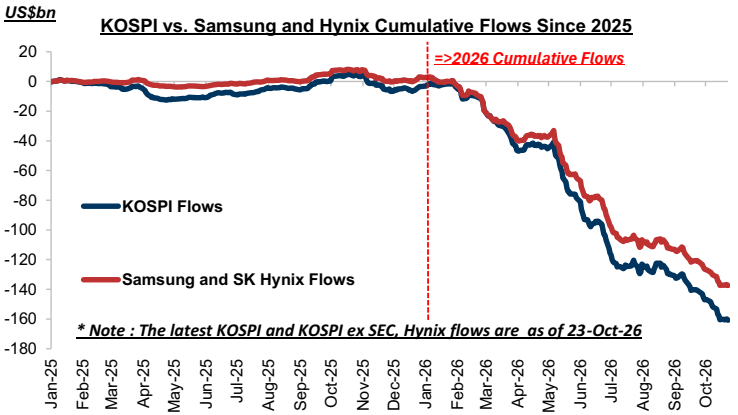
Charts of the week: Foreign Flows and Ownership Pulse Check

Exhibit 16: Corporate buybacks have been the only meaningful source of inflows in recent weeks, while all major investor groups have been net sellers



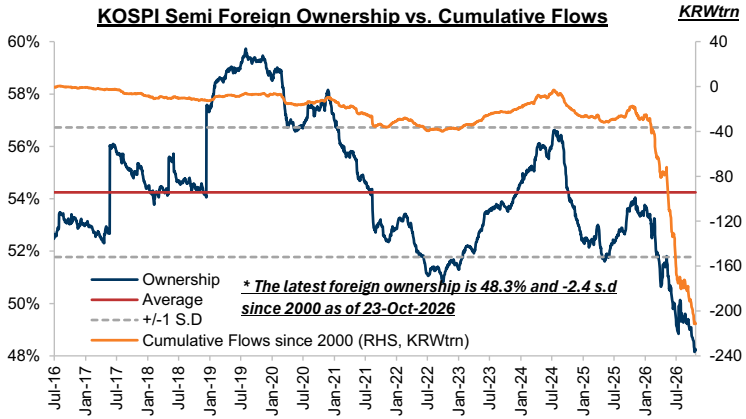
Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 17: Foreign investors have led year-to-date outflows, primarily driven by selling in the two largest memory stocks



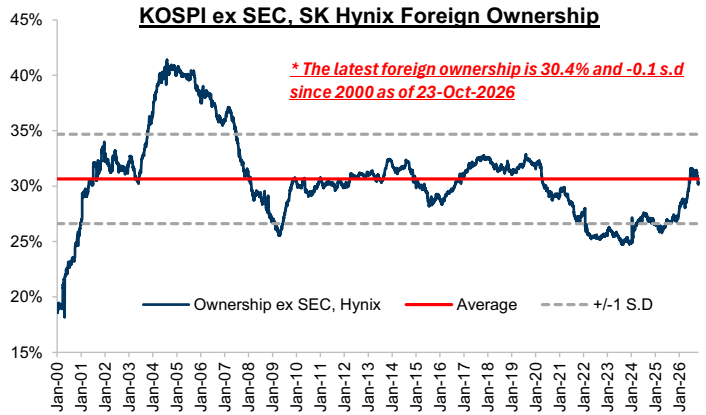
Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 18: Consequently, foreign ownership in the semiconductor sector has fallen to approximately -2.5 standard deviations relative to its historical range



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 19: Foreign ownership in the KOSPI excluding Samsung Electronics and SK Hynix has also declined below its long-term average



Source: Quantiwise, Goldman Sachs Global Investment Research

Summary

Exhibit 20: Summary of the past week's performance

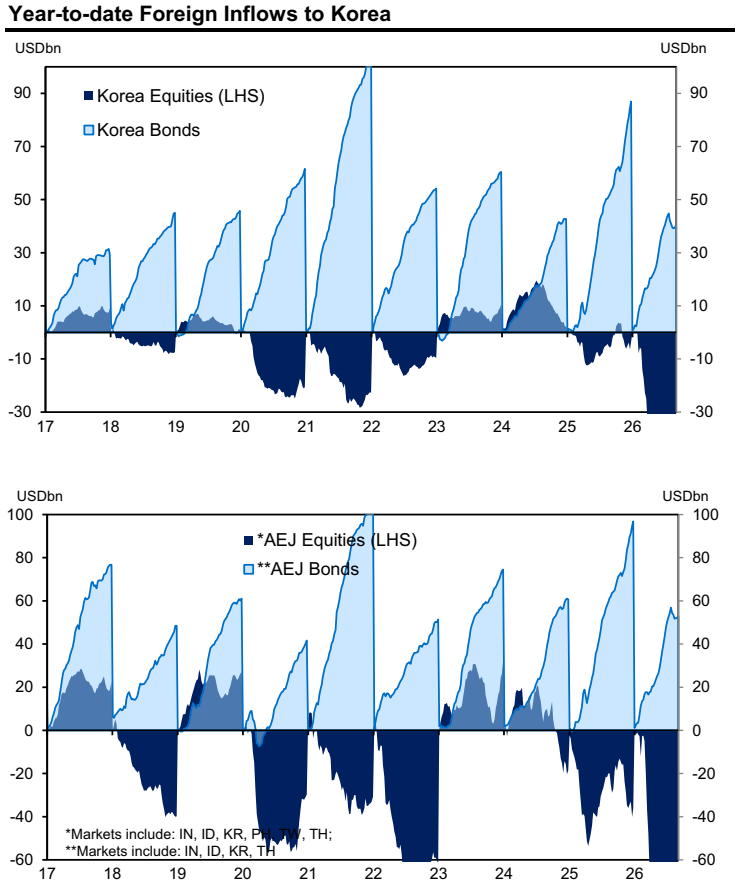
Equity Market Performance				
	P/E 2027E			1-wk chg
KOSPI	4.0	5,550.30	↓	(2.1)
KOSDAQ	17.1	751.69	↓	(5.4)
MSCI Korea	3.4	2,014.68	↓	(1.5)
KOSPI Performance by sector				
Top 3	wk chg (%)	**Bottom 3**		1wk chg (%)
Pharmaceuti	5.4	Chemicals		(9.4)
Software	2.3	Steel		(9.4)
Insurance	1.3	Shipbuilding		(7.9)
Market Flows				
Equities (KRW bn)		1-wk flow		in std dev*
KOSPI Flows: Foreigners		(709)	↓	*-0.1*
Institutions		422	↑	*0.2*
Individuals		(1,115)	↓	*-0.2*
Pensions		259	↑	*0.8*
KOSDAQ Flows: Foreigners		(675)	↓	*-1.0*
Institutions		(48)	↓	*0.0*
Individuals		684	↑	*0.4*
Pensions		(73)	↓	*-0.7*

FX/Interest Rate			
	Current		1-wk chg
USDKRW	1,325	↓	(0.7)
JPYKRW	8.47	↑	0.3
USDKRW 1M Risk Reversal/ATM vc	0.09	↑	2bp
USDKRW 1yr swap basis	(85)	↑	4bp
3-year KTB	3.80	↓	-5bp
10-year KTB	4.10	↓	-9bp

Up (↑) = Up wow vs. the previous week
Asterisk (*) = Expressed in standard deviation of 1-wk change in 1-year

Source: Bloomberg

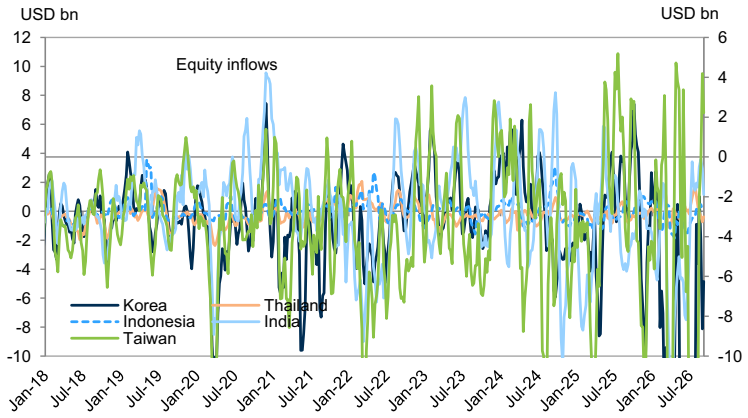
Exhibit 21: Summary of year-to-date flows



Source: Bloomberg

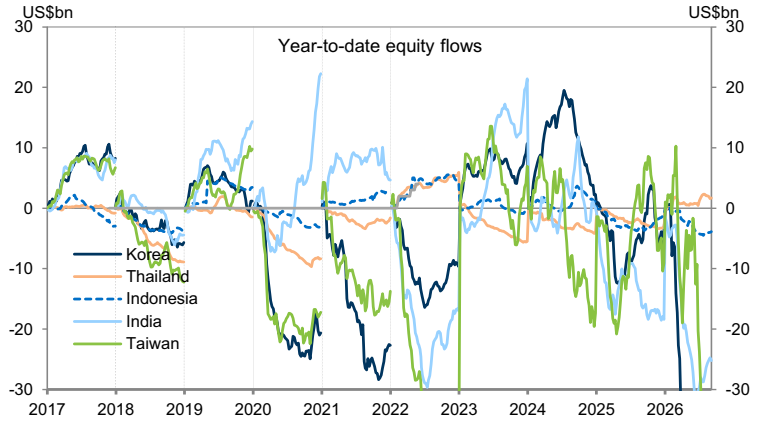
Investment flows

Exhibit 22: Equity inflows to 5 AEJ markets
4-week rolling sum



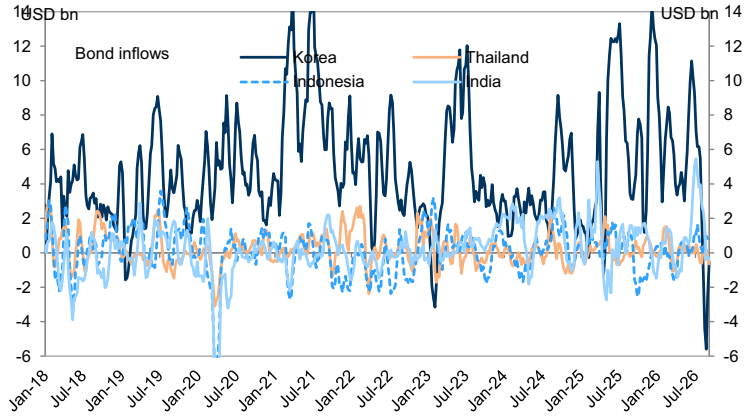
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 23: Equity inflows to 5 AEJ markets



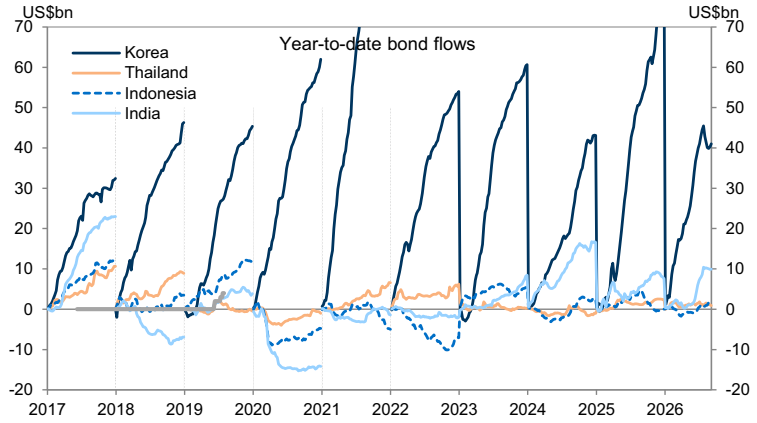
Source: Bloomberg

Exhibit 24: Bond inflows to 4 AEJ markets
4-week rolling sum



Source: Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

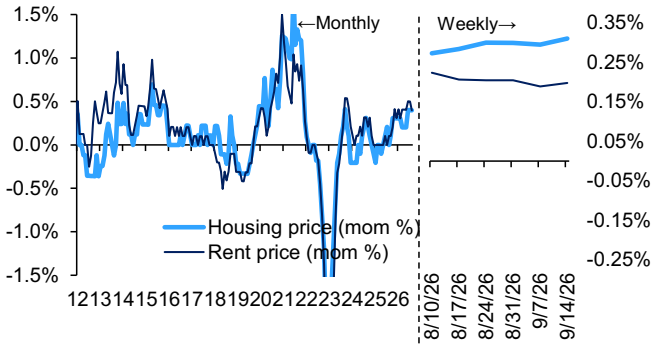
Exhibit 25: Bond inflows to 4 AEJ markets



Source: Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

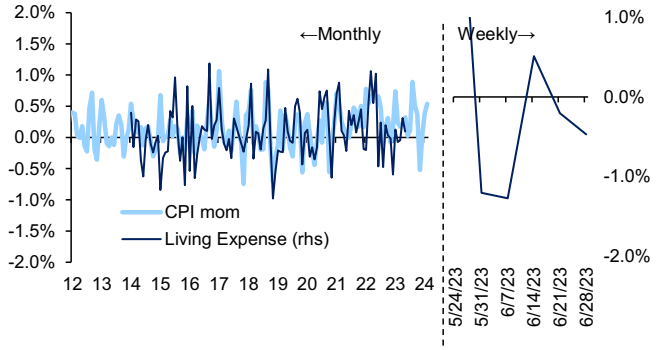
Macro Indicators

Exhibit 26: Housing prices and rental prices
National, monthly and weekly price changes



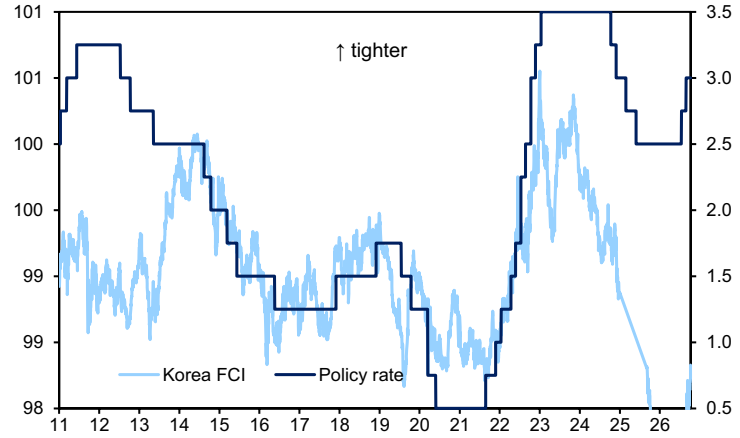
Source: KAB

Exhibit 27: Living expense price changes
Monthly and weekly changes vs. CPI



Source: Bloomberg, Korea Price Research Center, Goldman Sachs Global Investment Research

Exhibit 28: Daily Financial Condition Index
Jan 1, 2013=100



Source: Bloomberg, CEIC, Goldman Sachs Global Investment Research

Exhibit 29: Korea Economic Forecasts

Monthly indicators

Monthly (% yoy)	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Exports ex ships	29.0	52.8	47.9	54.4	72.9	64.0	75.2	86.0
Industrial production	-2.3	3.8	1.3	-1.1	6.2	4.0	-2.2	
Employment	0.8	0.7	0.3	-0.1	0.2	0.4	0.6	
CPI	2.0	2.2	2.6	3.1	3.2	2.8	3.1	2.9
BOK Policy Rate	2.50	2.50	2.50	2.50	2.50	2.75	3.00	3.00

Note: Blue-shaded cells denote positive weekly changes and acceleration in monthly changes

Macro and market outlook

Forecasts	2025	2026	2027	1Q26	2Q26	3Q26	4Q26	1Q27
Real GDP	1.1	3.4	2.6	3.8	3.8	2.9	3.3	2.2
Domestic Demand	0.6	2.4	2.5	2.2	2.3	2.4	2.6	1.7
Exports	4.3	10.7	4.8	11.8	8.9	9.8	12.2	6.4
Imports	3.3	8.8	5.9	8.5	6.2	9.1	11.3	8.0
Consumer Prices	2.1	2.6	2.1	2.1	3.0	3.0	2.5	2.5
Current Account - % GDP	6.5	18.1	18.2	14.7	21.0	19.2	17.0	15.8
KOSPI (end / 12mf)	4214	9000*						
Policy rate (end)	2.50	3.00	3.25	2.50	2.50	3.00	3.00	3.25
USDKRW (end)	1,444	1,433	1,364	1,346	1,501	1,419	1,348	1,382

* KOSPI for 2025 is the year-end level and KOSPI for 2026 is 12-month target

Source: Bloomberg, Goldman Sachs Global Investment Research

Performance

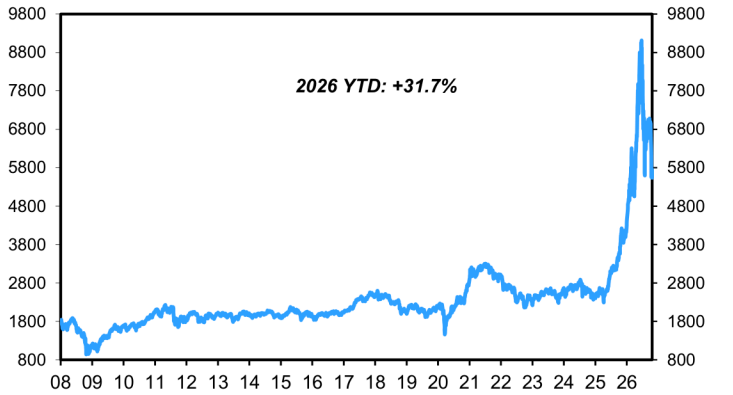
Exhibit 30: Performance of Korean and regional equity markets

		Index	Price return (%)					% vs average	
			Local Crncy		USD				
Korea	**Crn**	**23-Oct**	**1-wk**	**YTD**	**1-wk**	**YTD**	**RSI**	**30D**	**60D**
Kospi	KRW	5,550	(2.1)	31.7	(1.4)	43.1	31	(15)	(15)
Kospi 200	KRW	863	(1.8)	42.4	(1.1)	54.7	32	(16)	(16)
Kosdaq	KRW	752	(5.4)	(18.8)	(4.8)	(11.7)	36	(9)	(8)
MSCI Korea	KRW	2,015	(1.5)	46.0	(0.8)	58.7	30	(17)	(27)
Selected sector indexes									
Auto	KRW	2,361	(4.4)	(6.7)	(3.7)	1.4	34	(8)	(14)
Tech	KRW	87,266	(0.9)	75.3	(0.2)	90.5	34	44	3
Retail	KRW	521	(1.9)	8.6	(1.2)	18.0	33	(10)	(13)
Pharmaceutical	KRW	15,708	5.4	(9.1)	6.2	(1.3)	57	4	1
Steel	KRW	5,355	(9.4)	(18.1)	(8.8)	(11.0)	28	(16)	(19)
Chemicals	KRW	4,202	(9.4)	2.8	(8.8)	11.7	34	(11)	(10)
F&B	KRW	4,793	(1.2)	6.0	(0.5)	15.2	51	1	(0)
Machinery	KRW	2,374	(7.4)	(12.7)	(6.8)	(5.2)	28	(20)	(20)
Construction	KRW	135	(0.5)	36.9	0.2	48.7	32	(15)	(14)
Shipbuilding	KRW	1,112	(7.9)	(23.3)	(7.2)	(16.6)	28	(19)	(21)
Utilities	KRW	926	(0.4)	(33.7)	0.3	(27.9)	38	(5)	(9)
Banking	KRW	1,467	(4.6)	12.5	(3.9)	22.3	26	(11)	(11)
Securities	KRW	4,100	(4.0)	(2.2)	(3.4)	6.3	29	(13)	(17)
Insurance	KRW	41,812	1.3	37.1	2.0	49.0	37	(8)	(10)
Telecom	KRW	531	(6.4)	8.7	(5.7)	18.1	23	(10)	(12)
Software	KRW	191	2.3	(8.8)	3.0	(0.9)	44	(33)	(33)
Leisure	KRW	284	(2.4)	(43.7)	(1.7)	(38.8)	48	(54)	(54)
MSCI standard market and regional indexes									
AC Asia Pacific ex Japan	USD	811	(2.5)	12.3	(2.5)	12.3	29	(7)	(7)
USA	USD	7,300	0.0	11.8	0.0	11.8	45	(1)	(1)
EU	EUR	214	(1.0)	8.1	(1.2)	3.2	50	0	(1)
China	HKD	67	(5.7)	(19.6)	(6)	(20.2)	22	(7)	(10)
Japan	JPY	2,445	(1.7)	17.0	(0.9)	16.9	40	(3)	(3)
Regional indexes									
America									
S&P 500 (US)	USD	7,663	0.1	11.9	0.1	11.9	45	(1)	(0)
NASDAQ (US)	USD	26,009	(0.5)	11.9	(0.5)	11.9	39	(3)	(2)
BOVESPA (Brazil)	BRL	204,120	(0.6)	26.7	(0.3)	37.5	62	5	10
Asia									
Hang Seng (Hong Kong)	HKD	23,093	(4.5)	(9.9)	(4.5)	(10.6)	27	(5)	(7)
Hang Seng Ent. (China)	HKD	7,756	(4.6)	(13.0)	(4.5)	(13.6)	32	(5)	(7)
Topix (Japan)	JPY	3,966	(2.5)	16.3	(1.4)	16.6	38	(3)	(3)
TW Weighted (Taiwan)	TWD	46,041	(2.3)	59.0	(2.8)	53.5	40	(3)	(0)
Straits Times(Singapore)	SGD	5,422	0.1	16.7	0.3	17.4	36	(3)	(4)
S&P CNX Nifty (India)	INR	22,600	(2.4)	(13.5)	(2.3)	(18.1)	42	(1)	(4)
Europe									
DJ EURO STOXX 50 (EU)	EUR	6,038	(2.7)	4.3	(2.9)	(0.5)	39	(3)	(5)
FTSE 100 (UK)	GBP	10,660	(2.0)	7.3	(1.6)	6.6	53	0	(1)

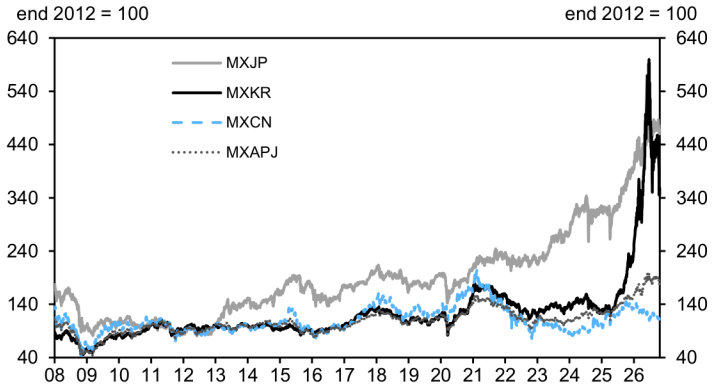
Source: Bloomberg, MSCI

Exhibit 31: KOSPI and MSCI regional index

KOSPI Index price performance (KRW)

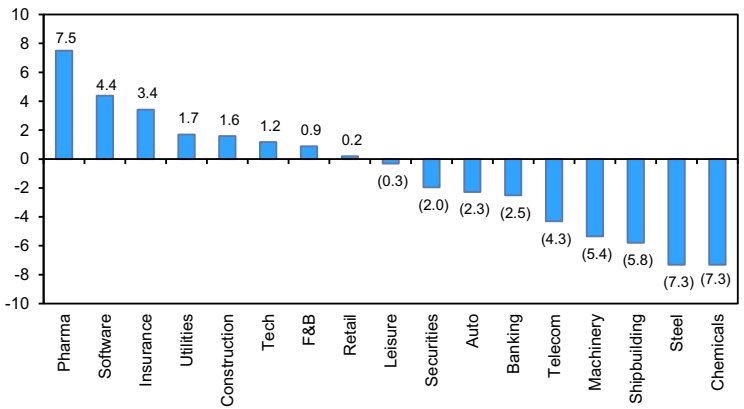


Regional MSCI Index performance



Source: Bloomberg, MSCI

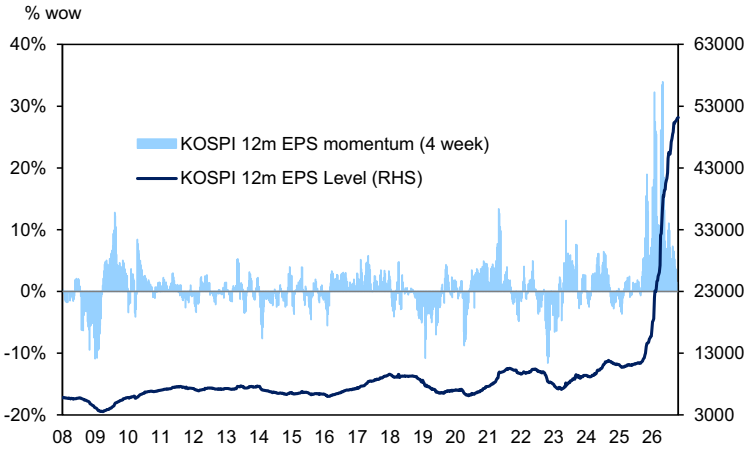
Exhibit 32: Weekly sector performance relative to KOSPI



Note: Return relative to KOSPI = price return of the sector (1-wk) – price return of KOSPI (1-wk).

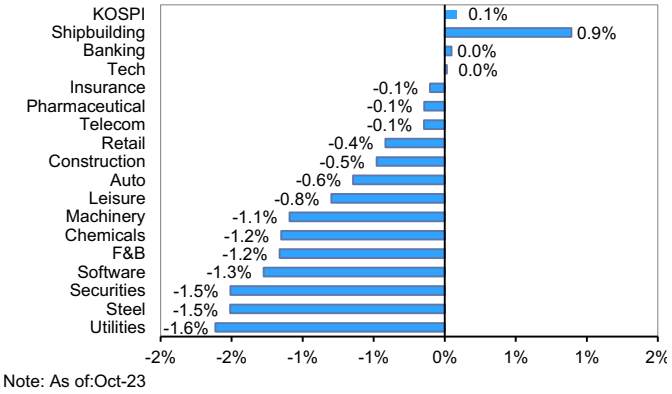
Source: Bloomberg, Quantiwise

Exhibit 33: Historical trend of monthly KOSPI earnings momentum



Source: Quantiwise

Exhibit 34: Weekly earnings momentum for KOSPI and selected sectors



Note: As of:Oct-23

Source: Quantiwise

Exhibit 35: KOSPI earnings revisions by sector

As of: Oct-23		2026E EPS Revision		2027E EPS Revision	
Sector	**Index wgt**	**6M**	**3M**	**6M**	**3M**
Tech	21%	**35%**	**4%**	**55%**	**3%**
Software	6%	-13%	-13%	-10%	-14%
Chemicals	12%	**104%**	**14%**	**28%**	**14%**
Auto	8%	-7%	-4%	-11%	-7%
Steel	4%	-1%	**8%**	-5%	**3%**
Machinery	2%	-9%	-10%	-12%	-15%
Shipbuilding	1%	**7%**	**4%**	**7%**	**3%**
Retail	7%	**11%**	**5%**	**14%**	**15%**
F&B	3%	-1%	-6%	-2%	-4%
Pharmaceutical	2%	**2%**	**16%**	-4%	**15%**
Leisure	1%	-21%	-21%	-15%	-21%
Utilities	4%	-44%	-22%	-38%	-26%
Telecom	2%	**2%**	**3%**	**3%**	**1%**
Construction	2%	**23%**	-2%	**15%**	0%
Insurance	5%	**23%**	**12%**	**37%**	**19%**
Banking	5%	-6%	-10%	-6%	-9%
Securities	2%	**27%**	-3%	**6%**	-8%
KOSPI		**34%**	**7%**	**46%**	**5%**

Source: Quantiwise

Valuations

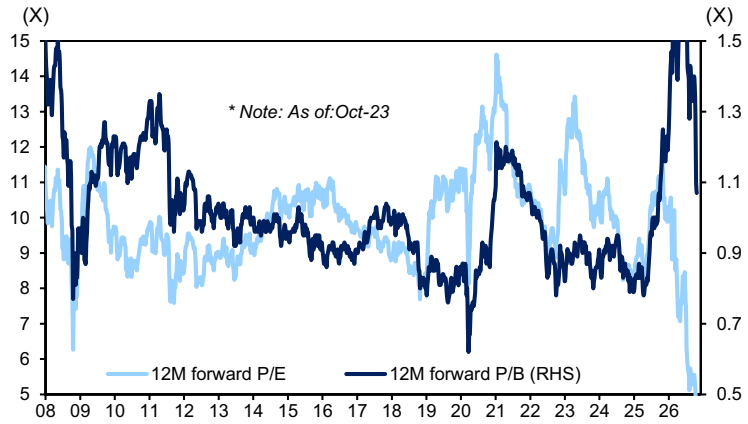
Exhibit 36: Valuation summary for regional equity markets

			Consensus estimates													
	Local Currency	Bloomberg Ticker	P/E (X)		EPS Growth (%)		P/B (X)		D/Y (%)		ROE (%)			EBIT Margin (%)		P/CF (%)
			2027E	2028E	2027E	2028E	2027E	2028E	2027E	2028E	2026E	2027E	2028E	2027E	2028E	2027E
MSCI markets																
Australia	AUD	MXAU	17.9	17.4	4.9	3.1	2.5	2.4	3.7	3.8	14.2	14.4	14.2	27.7	27.2	10.8
Singapore Free	SGD	SIMSCI	15.9	14.4	10.9	9.8	2.0	1.9	4.0	4.1	12.3	13.0	13.6	24.4	24.8	6.8
Hong Kong	HKD	MXHK	13.3	12.4	4.3	7.4	1.1	1.1	4.3	4.5	8.4	8.4	8.7	14.6	14.5	10.3
India	INR	MXIN	18.0	15.6	16.0	15.1	2.7	2.4	1.6	1.8	15.0	15.6	16.0	18.0	18.8	12.9
Indonesia	IDR	MXID	7.5	6.9	9.1	9.0	1.3	1.1	9.0	9.6	16.5	17.1	16.6	32.2	33.1	3.9
Japan	JPY	MXJP	14.5	12.9	14.6	12.1	1.7	1.6	2.3	2.5	11.8	12.5	13.0	13.1	14.2	10.4
Korea	**KRW**	**MXKR**	**3.6**	**3.4**	**38.7**	**5.2**	**1.2**	**0.9**	**3.9**	**4.9**	**38.0**	**37.4**	**29.7**	**43.9**	**41.6**	**3.1**
Malaysia	MYR	MXMY	13.0	12.0	6.2	7.3	1.4	1.3	4.9	5.2	10.7	10.9	11.2	26.5	27.6	8.4
Pakistan	PKR	MXPK	6.0	5.5	8.3	9.4	1.2	1.3	7.6	9.2	20.8	20.2	23.3	24.5	27.1	22.1
Philippines	PHP	MXPH	9.4	8.6	11.2	9.2	1.3	1.2	3.7	3.9	14.9	15.1	14.9	34.3	32.0	
Taiwan	TWD	MXTW	16.1	12.6	30.8	27.8	4.2	3.4	2.2	2.7	27.7	29.0	30.2	17.3	17.6	8.2
Thailand	THB	MXTH	16.2	14.9	2.6	5.9	1.9	1.8	3.8	4.1	12.5	12.2	12.2	12.7	12.5	11.6
AC Asia Pacific	USD	MXAP	10.1	8.9	28.0	12.7	1.7	1.5	2.8	3.2	16.8	18.3	17.9	23.0	22.8	
AC Asia Pacific ex Japan	USD	MXAPJ	8.9	8.0	31.3	11.9	1.7	1.5	3.0	3.5	19.4	21.2	20.1	27.2	26.2	14.2
AC Asia ex Japan	USD	MXASJ	8.3	7.4	33.6	12.5	1.7	1.4	3.0	3.5	20.1	21.9	20.7	27.2	26.2	8.0
AC Far East ex Japan	USD	MXFEJ	7.7	6.8	35.2	12.4	1.6	1.3	3.2	3.7	20.6	22.4	21.1	27.9	26.8	0.0

Note: As of Oct-22-2026

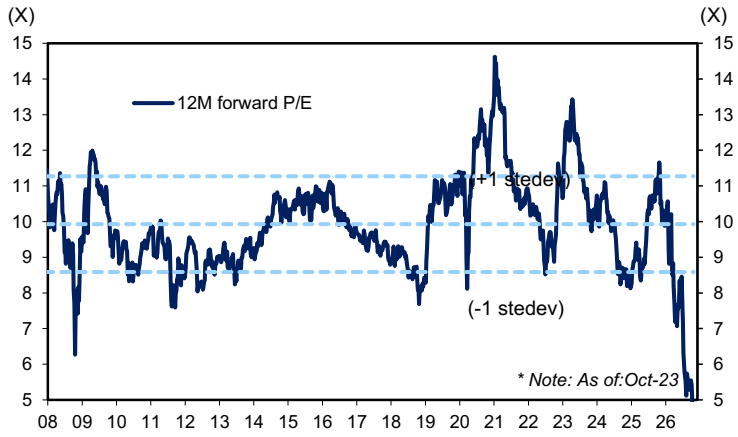
Source: Bloomberg, I/B/E/S, FactSet, Goldman Sachs Global Investment Research

Exhibit 37: 12-month forward P/E and P/B for KOSPI, since Jan 2008



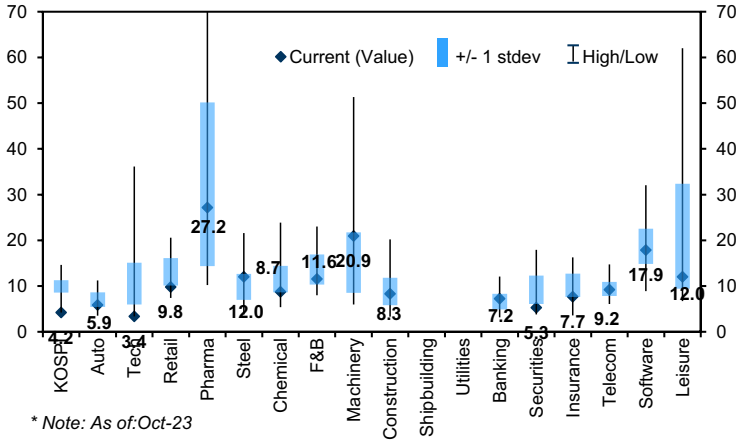
Source: Quantiwise

Exhibit 38: 12-month forward P/E for MSCI Korea, since Jan 2008



Source: Quantiwise

Exhibit 39: Current 12m forward P/E of KOSPI and selected sectors since Jan 2006



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 40: 12-m forward P/E for KOSPI and selected sectors since Jan 2006

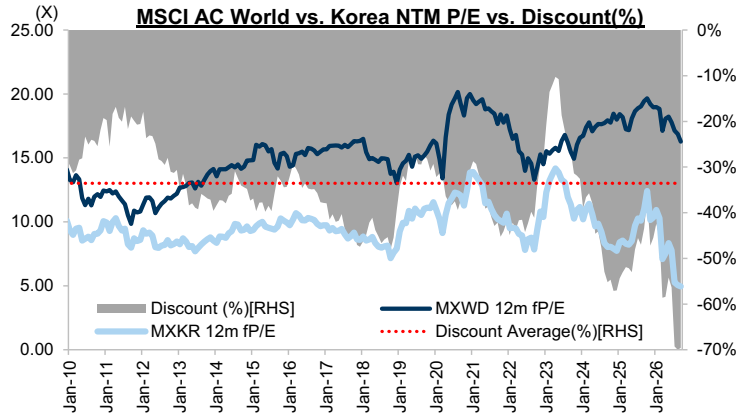
KOSPI sector	Current	Low	-1std.dev	Mean	+1std.dev	High
KOSPI Index	**4.2**	4.2	8.6	9.9	11.3	14.6
Auto	**5.9**	3.5	5.4	7.0	8.6	11.2
Tech	**3.4**	3.4	6.0	10.5	15.1	36.2
Retail	**9.8**	7.4	10.1	13.1	16.1	20.6
Pharmaceuticals	**27.2**	10.2	14.4	32.3	50.2	72.8
Steel	**12.0**	3.8	7.0	9.8	12.6	21.6
Chemical	**8.7**	5.4	8.6	11.5	14.4	23.9
F&B	**11.6**	8.0	10.3	13.6	16.9	23.0
Machinery	**20.9**	6.0	8.5	15.1	21.7	51.3
Construction	**8.3**	3.3	5.8	8.8	11.8	20.2
Shipbuilding	**17.1**			21.5		
Banking	**7.2**	3.2	4.9	6.6	8.3	12.1
Securities	**5.3**	3.8	6.1	9.2	12.3	17.9
Insurance	**7.7**	3.6	7.6	10.2	12.7	16.3
Telecom	**9.2**	6.1	7.8	9.4	10.9	14.7
Software	**17.9**	8.9	14.8	18.7	22.5	32.0
Leisure	**12.0**	6.7	9.4	20.9	32.4	62.0

Note: As of:Oct-23

Source: Quantiwise

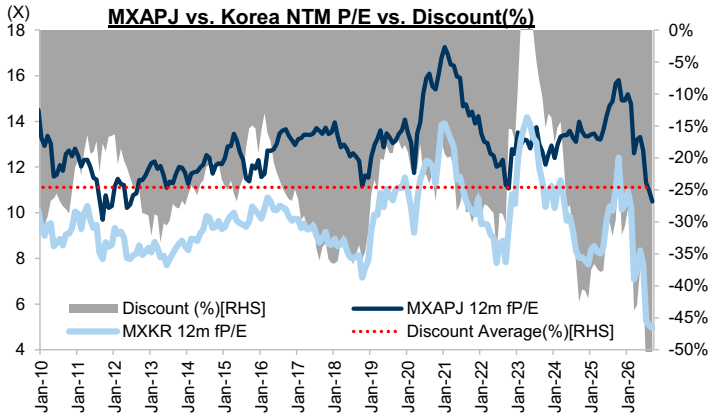
Valuation discount relative to Global and Asia regional peers

Exhibit 41: MXKR valuation discount relative to MSCI AC World NTM P/E



Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 42: MXKR valuation discount relative to MXAPJ NTM P/E



Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 43: NTM P/E Valuation Discount vs. 10-year History by Sector

NTM P/E Valuation Discount vs. 10-Year History				
	Current		Z-Score	
	vs. MXWD	vs. MXAPJ	vs. MXWD	vs. MXAPJ
Energy	-39%	-33%	-0.66	-0.53
Materials	-19%	-17%	-0.02	-0.40
Industrials	-63%	-35%	-2.75	-2.45
Consumer Discretionary	-63%	-46%	-0.10	0.92
Consumer Staples	-18%	-16%	0.29	1.15
Health Care	71%	8%	-1.31	-1.22
Financials	-37%	-29%	0.95	1.03
Information Technology	-76%	-52%	-1.21	-0.94
Communication Services	-25%	-1%	-0.97	0.25
Utilities	-	-	-	-
MXKR	**-70%**	**-53%**	**-2.75**	**-2.42**

Source: FactSet, MSCI, Goldman Sachs Global Investment Research

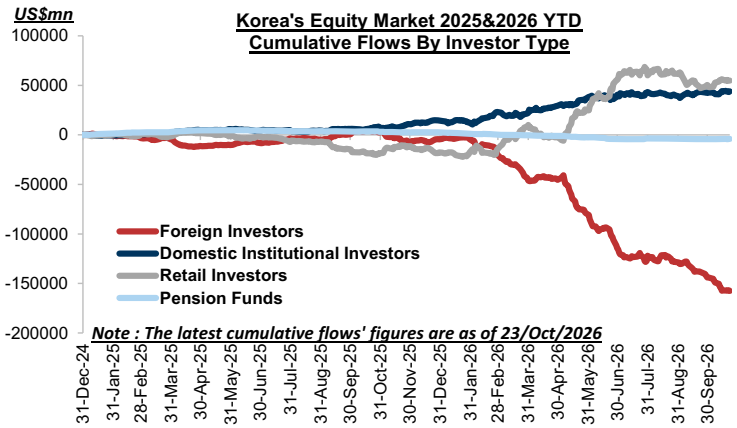
Exhibit 44: P/B Valuation Discount vs. 10-year History by Sector

P/B Valuation Discount vs. 10-Year History				
	Current		Z-Score	
	vs. MXWD	vs. MXAPJ	vs. MXWD	vs. MXAPJ
Energy	-46%	-4%	-0.30	1.38
Materials	-82%	-80%	-2.32	-2.35
Industrials	-51%	6%	1.15	1.32
Consumer Discretionary	-80%	-61%	-0.12	0.98
Consumer Staples	-38%	-16%	1.39	1.81
Health Care	-18%	-14%	-1.22	-1.55
Financials	-49%	-32%	0.84	2.51
Information Technology	-65%	-33%	0.94	1.00
Communication Services	-73%	-55%	-1.41	-1.01
Utilities	-80%	-74%	0.19	0.45
MXKR	**-38%**	**-4%**	**1.89**	**2.43**

Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Flows

Exhibit 45: 2025 & 2026 year-to-date KOSPI purchase by investor type



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 46: KOSPI net equity transactions by investor type (KRW bn)

	Weekly			
	Sep 25 - Oct 02	Oct 01 - Oct 08	Oct 09 - Oct 16	Oct 16 - Oct 23
Financial Institutions	(972)	(2,389)	3,418	422
- Banks	(2)	(2)	(17)	(12)
- Securities	(383)	(2,231)	3,993	(37)
- Insurance	(13)	(44)	498	339
- Investment trusts	(275)	317	(744)	111
- Pension funds	(156)	(373)	79	259
- Other financial institutions	78	35	51	(1)
- PEF	(222)	(91)	(442)	(237)
Foreigners	(8,323)	(6,469)	(11,239)	(709)
Individuals	1,902	6,360	4,312	(1,115)
Others	7,385	2,482	3,476	1,402
(YTD)				
Financial Institutions	38,414	36,025	39,443	39,865
- Investment trusts	(6,543)	(6,226)	(6,970)	(6,859)
- Pension funds	(8,970)	(9,343)	(9,264)	(9,005)
Foreigners	(190,770)	(197,239)	(208,479)	(209,187)
Individuals	91,295	97,654	101,966	100,851

Source: Quantiwise, Goldman Sachs Global Investment Research

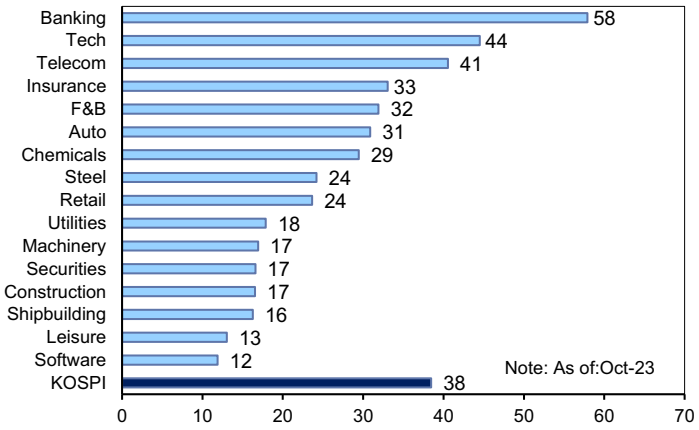
Exhibit 47: Weekly fund flows on sectors by investor type, with relative returns to KOSPI

KOSPI and sector's relative returns and flow data

Sectoral flows by investor type	Soft	Ins	Util	Cons	Tech	F&B	Retail	Leis	Sec	Auto	Bank	Telco	Mach	Ship	Steel	Chem	KOSPI
Rel return to KOSPI (%)	4.4%	3.4%	1.7%	1.6%	1.2%	0.9%	0.2%	-0.3%	-2.0%	-2.3%	-2.5%	-4.3%	-5.4%	-5.8%	-7.3%	-7.3%	**-2.1%**
Foreign Flows / Mkt Cap (bp)	-2	-7	1	25	-3	-19	-7	-22	-3	-3	-7	-6	-8	0	3	5	**-2**
Retail Flows / Mkt Cap (bp)	-13	-1	-8	-11	-1	-2	-2	-5	1	-1	2	3	16	6	6	6	**-2**
Pension Flows / Mkt Cap (bp)	14	8	0	-10	0	8	5	-5	-2	3	-1	1	-2	-1	-7	-7	**1**
Inv Trust Flows / Mkt Cap (bp)	5	1	1	2	0	12	1	8	1	2	0	0	-5	-3	-3	-1	**0**
Foreign Flows (KRW bn)	-12	-70	3	76	-742	-91	-60	-18	-13	-59	-134	-21	-65	-13	18	75	**-709**
Retail Flows (KRW bn)	-79	-15	-20	-33	-374	-10	-17	-4	5	-14	31	12	136	217	40	94	**-1115**
Pension Flows (KRW bn)	83	80	-1	-30	31	38	45	-4	-11	43	-22	3	-20	-52	-47	-110	**259**
Inv Trust Flows (KRW bn)	31	7	2	7	-22	60	13	7	6	33	0	-2	-45	-104	-24	-15	**111**

Source: Quantiwise, Korea Exchange, Goldman Sachs Global Investment Research

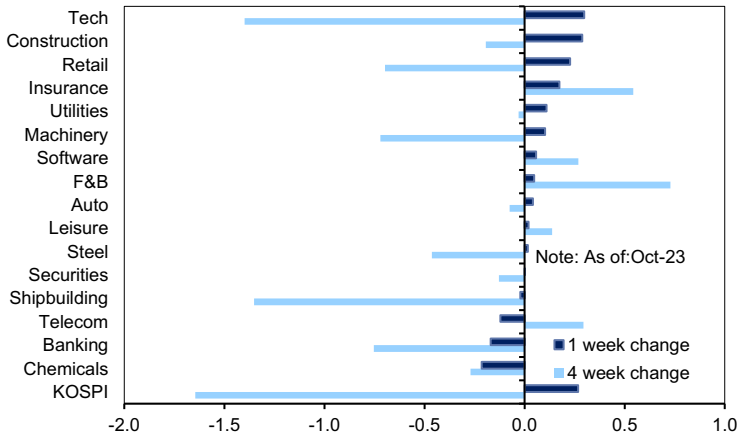
Exhibit 48: KOSPI sectors, ranked by foreign ownership as a % of total market value



Source: Korea Exchange, Quantiwise

Exhibit 49: KOSPI sectors ranked by foreign ownership % change

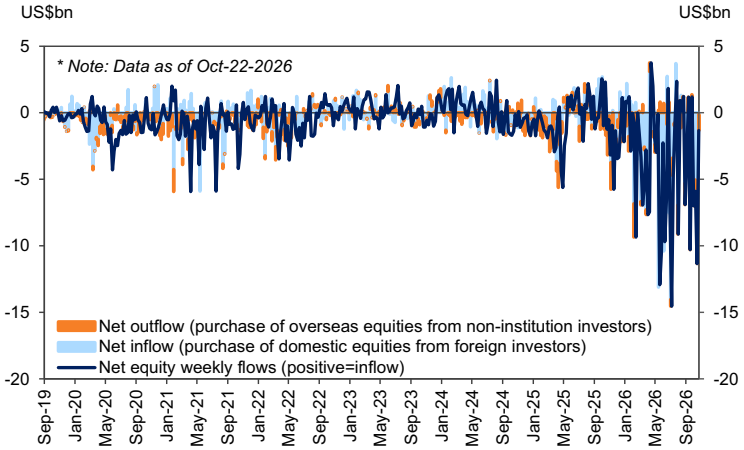
Absolute ownership % (market value) changes



Source: Korea Exchange, Quantiwise

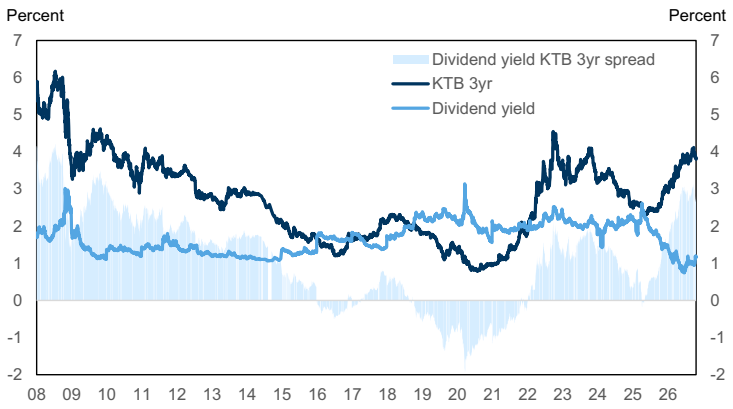
Exhibit 50: Equity net weekly flows

(Week ending on Thursday due to data availability)



Source: KSD, Quantiwise, Haver, Goldman Sachs Global Investment Research

Exhibit 51: 12-month forward D/Y for KOSPI and Korea Treasury bond yield



Source: Bloomberg, Goldman Sachs Global Investment Research

Currency, rates and commodities

Exhibit 52: Currency, rates and commodities

Sectoral flows by investor type

Currency	Bloomberg Ticker	Current	Price Change (%) 1-wk	1-mo	3-mo	YTD
US Dollar/ Korean Won	USDKRW	1,325	(0.7)	(3.1)	(10.0)	(8.0)
US Dollar/ Korean Won - 3M NDF	KWN+3M	1,321	(0.8)	(3.1)	(10.3)	(8.0)
US Dollar/ Korean Won - 6M NDF	KWN+6M	1,318	(0.8)	(3.2)	(10.3)	(7.9)
US Dollar/ Korean Won - 12M NDF	KWN+12M	1,310	(0.8)	(3.3)	(10.5)	(8.0)
Japanese Yen / Korean Won	JPYKRW	8.47	0.3	(1.8)	(5.9)	(8.1)
Euro/ Korean Won	EURKRW	1,485	(1.0)	(4.5)	(11.5)	(12.4)
Chinese Yuan/ Korean Won	CNYKRW	198.6	(0.5)	(2.5)	(8.8)	(3.9)
Euro/ US Dollar	EURUSD	1.121	(0.3)	(1.5)	(1.5)	(4.6)
US Dollar/ Chinese Yuan	USDCNY	6.670	(0.2)	(0.6)	(1.6)	(4.5)
US Dollar/ Japanese Yen	USDJPY	156.38	(1.0)	(1.2)	(4.6)	(0.2)

Fixed Income	Bloomberg Ticker	Yield (%) Last Close	-1 wk	-1 mo	-3 mo
Korea 3M CD	KSDACD3M	3.26	3.25	3.21	2.91
Korea Treasury Bond 3yr	KOTB3YR	3.80	3.85	4.01	3.92
Korea Treasury Bond 10yr	KOTB10YR	4.10	4.18	4.39	4.39

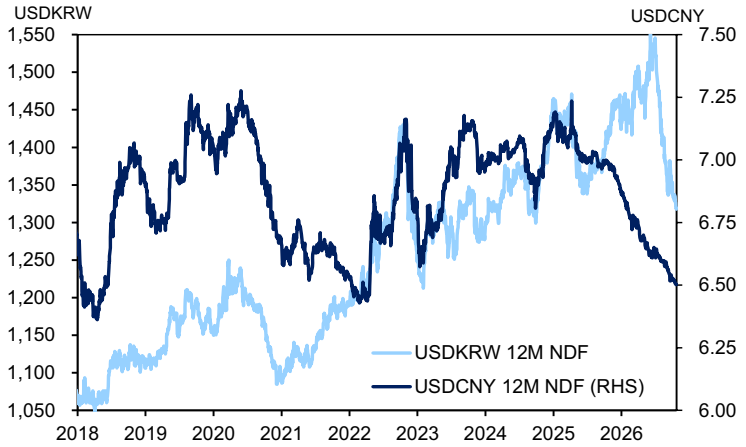
Major Commodity prices

Commodity	Bloomberg Ticker	Unit	Crncy	Last Close	Price Change (%) 1-wk	1-mo	3-mo	YTD
Baltic Dry	BDIY	N/A	USD	2,693	5.8	(21.5)	(0.8)	43.5
WTI	CL1	per barrel	USD	101	(0.2)	9.8	9.8	76.2
Coal	XAA	per ton	USD	152	(0.4)	12.1	21.5	57.9
Gold	GOLDS	per oz.	USD	4,027	(3.3)	(6.1)	(0.6)	(6.8)
Aluminum	LA1	per ton	USD	3,041	1.8	(6.9)	(4.8)	2.1
Copper	LP1	per ton	USD	14,144	(1.9)	(4.8)	2.3	13.5
Steel	STEEL	per ton	USD	3,267	1.2	3.8	(3.4)	28.4
Zinc	LMZSDY	per ton	USD	3,716	0.6	(6.2)	2.6	20.6
Corn	C1	per bushel	USD	460	(3.0)	(13.0)	(0.9)	4.5
Wheat	W1	per bushel	USD	627	(5.1)	(11.5)	(10.0)	23.6
Soybean	S1	cent per bushel	USD	1,255	(0.5)	(4.8)	1.4	21.8
DRAM*	ISPPDR29	per chip	USD	12.3	0.8	5.7	8.4	229.0
DRAM**	ISPPDR35	per chip	USD	18.6	4.3	12.4	40.4	304.8

Note: We show spot prices for * DDR3 4Gb 512Mx8 1333/1600MHz and ** DDR4 4Gb 542Mx8 2133/2400 MHz

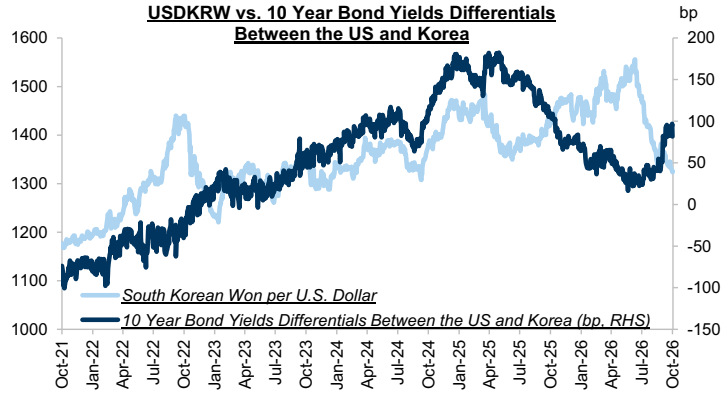
Source: Bloomberg

Exhibit 53: USDKRW and USDCNY rates



Source: Bloomberg

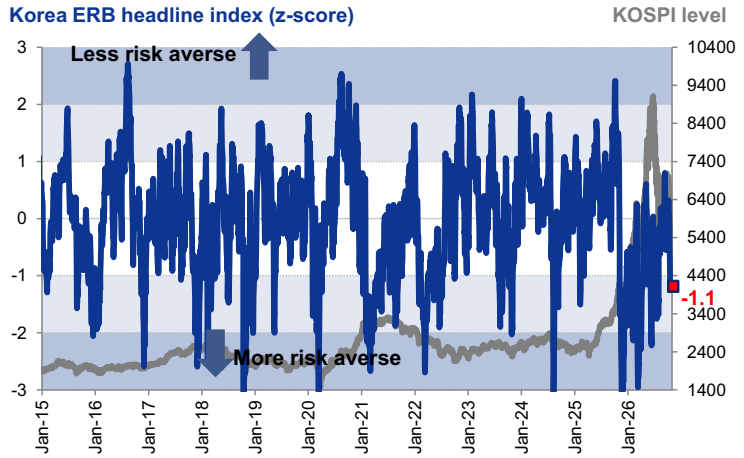
Exhibit 54: USDKRW vs. 10 year bond yields differentials between the US and Korea



Source: FactSet, Goldman Sachs Global Investment Research

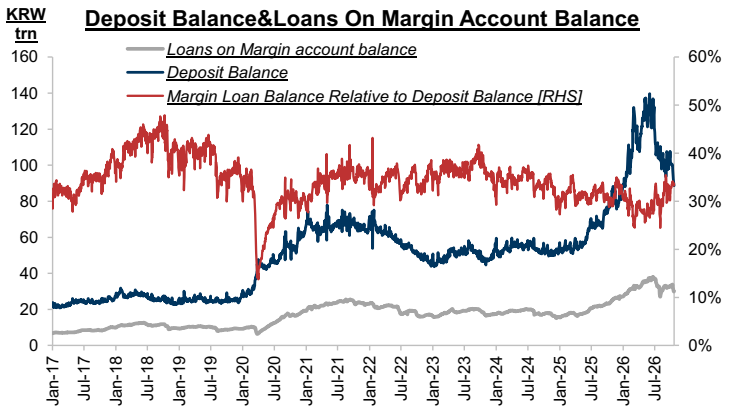
Korea ERB, Credit and Market Technicals

Exhibit 55: Korea Equity Risk Barometer



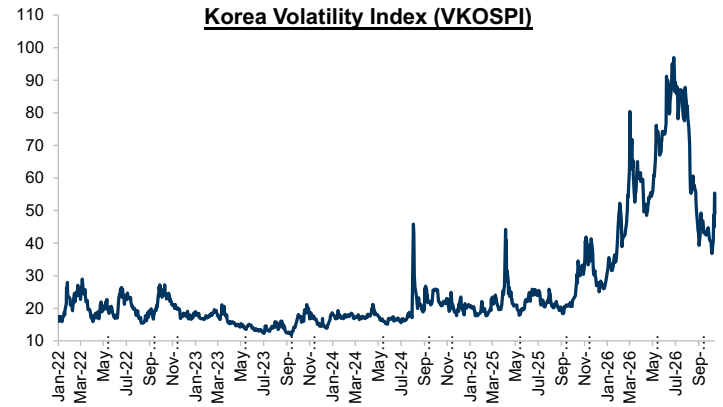
Source: Goldman Sachs Global Investment Research

Exhibit 56: Deposit Balance & Loans on Margin account balance



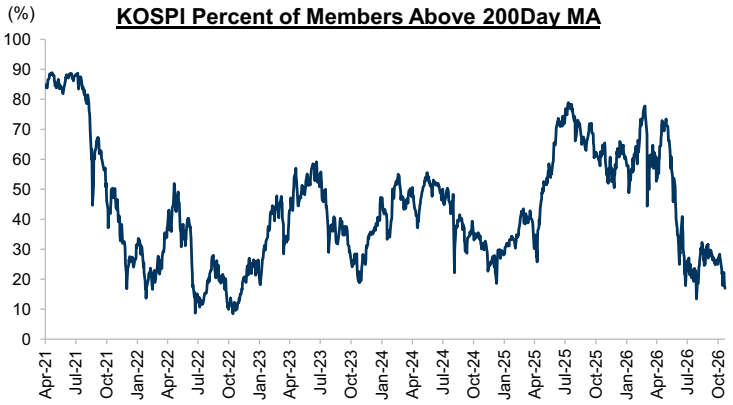
Source: Bloomberg, Korea Financial Investment Association, Goldman Sachs Global Investment Research

Exhibit 57: Korea Volatility Index (VKOSPI)



Source: Quantiwise, Goldman Sachs Global Investment Research

Exhibit 58: KOSPI Percent of Members above 200 day moving average



Source: Bloomberg, Goldman Sachs Global Investment Research

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Reg AC

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Contributing Authors: Timothy Moe, CFA Goldman Sachs (Singapore) Pte, John Kwon Goldman Sachs (Singapore) Pte.

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